

MOOSIC

A Music Clustering Case
Study: Segmenting Songs
Using Machine Learning



Project Overview

Objective

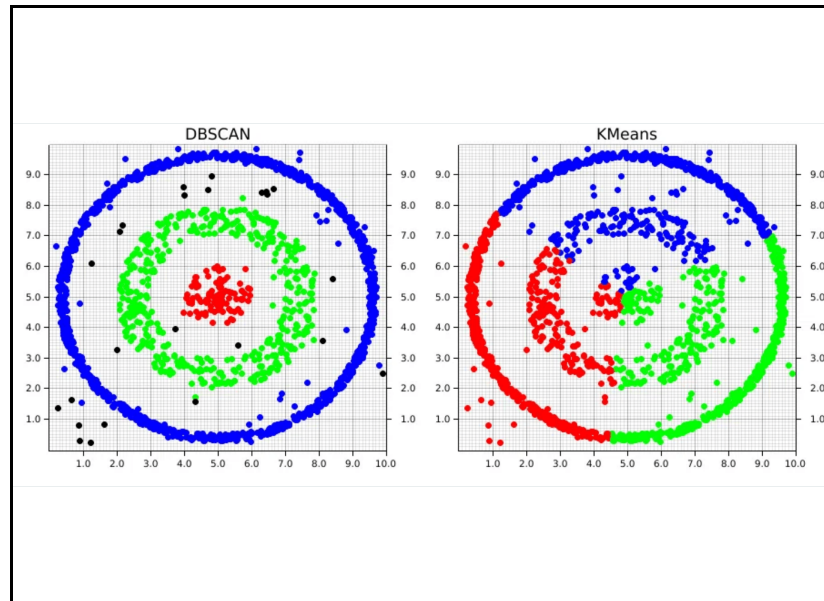
To group songs into distinct clusters based on their unique audio features using advanced machine learning algorithms.

Dataset

A curated collection of tracks with attributes like danceability, energy, and acousticness sourced from Spotify.

Goal

Enable data-driven music recommendation and automated playlist generation through precise segmentation.



Data & Preprocessing

Key Audio Features

Danceability & Energy

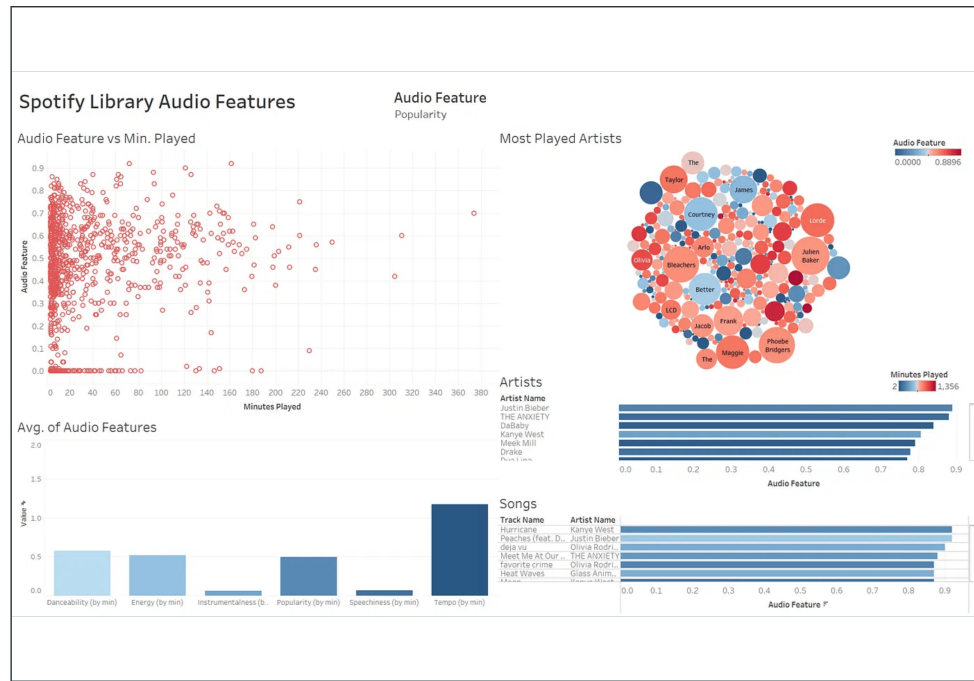
Acousticness & Valence

Instrumentalness & Tempo

Preprocessing Steps

Data cleaning, feature selection, and indexing by track identity.

Handling multi-dimensional variance through normalization.



Feature Scaling

The Challenge

Why Scale Data?

Features like 'Tempo' (e.g., 120 BPM) have much larger ranges than 'Danceability' (0-1). Without scaling, algorithms like KMeans would be biased toward high-magnitude features.

Method 01

Min-Max Scaling

Rescales all data points to a fixed range, typically $[0, 1]$. This preserves the relative relationships between data points while ensuring all features contribute equally.

Method 02

Standard Scaling

Transforms data to have a mean of 0 and a standard deviation of 1. This is particularly useful for algorithms that assume a Gaussian distribution of the data.

Finding Optimal Clusters

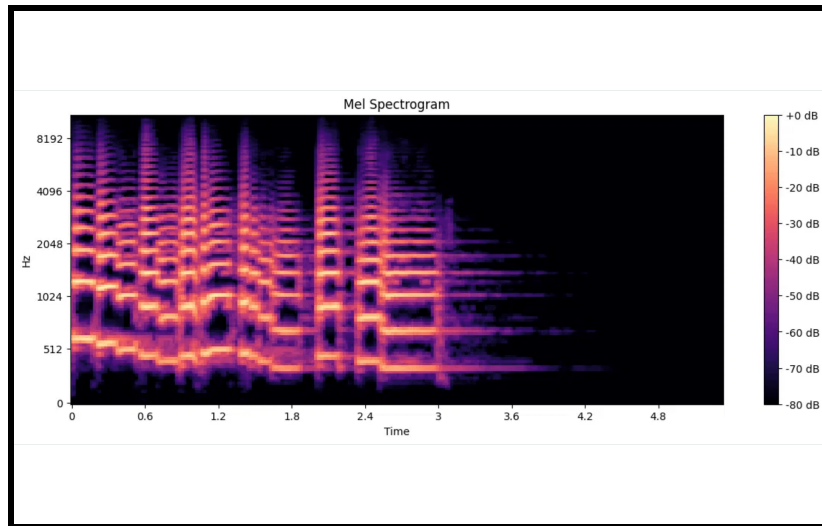
The Elbow Method

Analyzing Inertia scores to identify the "elbow" point—where the rate of decrease in within-cluster sum of squares significantly slows down.

Silhouette Score

Validating cluster quality by measuring how similar an object is to its own cluster compared to others, ensuring distinct segmentation.

Optimal Result: 7 Clusters



Visualizing Audio Complexity: Mel Spectrogram

KMeans: Cluster Analysis

Cluster Profiles

Party Tracks

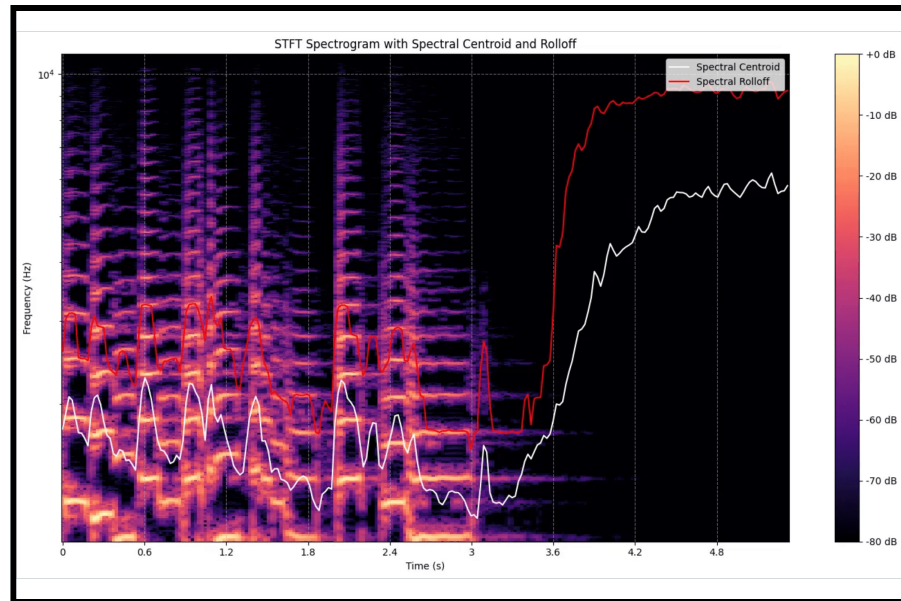
High Energy and Danceability. Characterized by strong rhythmic patterns and high tempo.

Acoustic / Classical

High Acousticness and Instrumentalness. Low energy profiles with complex harmonic structures.

Upbeat / Happy

High Valence and Tempo. Positive musical sentiment designed for mood elevation.



Technical Feature Extraction

DBSCAN Methodology

Density-Based Approach

Core Concept

Groups together points that are closely packed together, defining clusters as high-density regions separated by low-density areas.

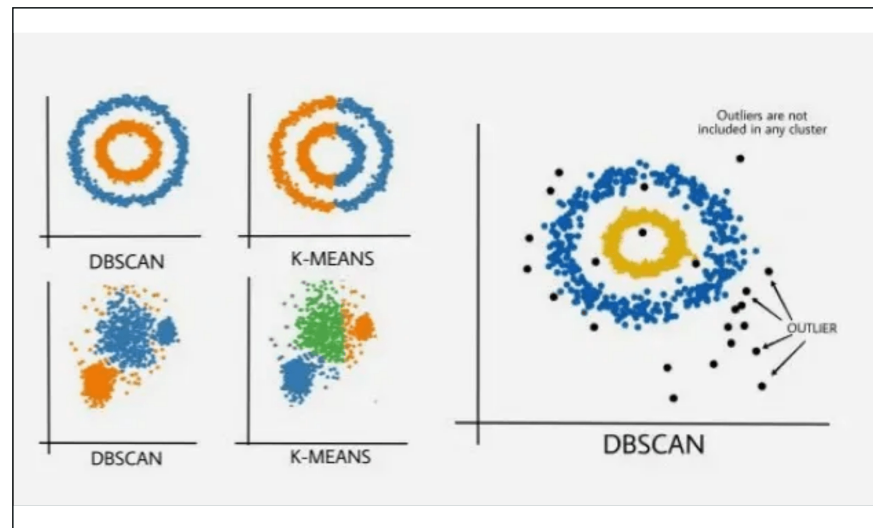
Epsilon Selection

Utilized the **k-Distance Graph** to identify the "elbow" point, determining the optimal distance threshold for neighborhood definition.

Noise Detection

Unlike KMeans, DBSCAN effectively identifies outliers as noise, preventing them from distorting cluster centroids.

DBSCAN vs. K-Means



Handling Arbitrary Shapes & Outliers

DBSCAN: Results & Noise

13

Clusters Found

17.59%

Noise Points

Model Configuration

Epsilon (ϵ): 0.14 — The maximum distance between two samples for one to be considered as in the neighborhood of the other.

Min Samples: 5 — The number of samples in a neighborhood for a point to be considered as a core point.

Key Insight

DBSCAN successfully identified high-density clusters while isolating nearly 18% of the dataset as noise. These "noise" tracks represent unique musical outliers that defy standard categorization, providing a valuable tool for identifying avant-garde or genre-bending music.

Agglomerative Clustering

Methodology

Bottom-Up Approach

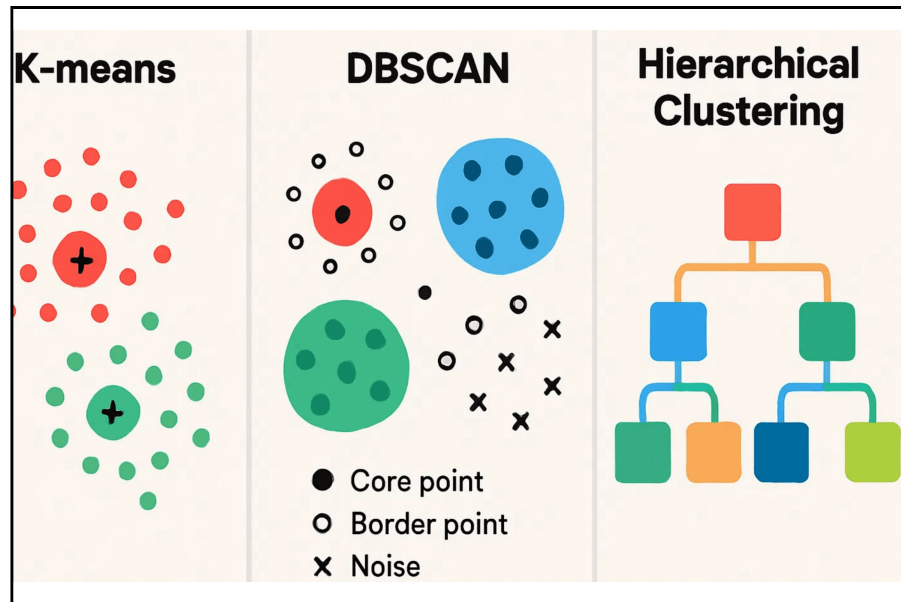
A hierarchical method where each observation starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.

Dendrogram Analysis

Visualized the hierarchy of song relationships to identify the natural groupings and decide on the optimal number of clusters.

Implementation

Segmented the data into **20 clusters** using Ward's linkage method for finer granularity in music recommendation.



Hierarchical Relationship Mapping

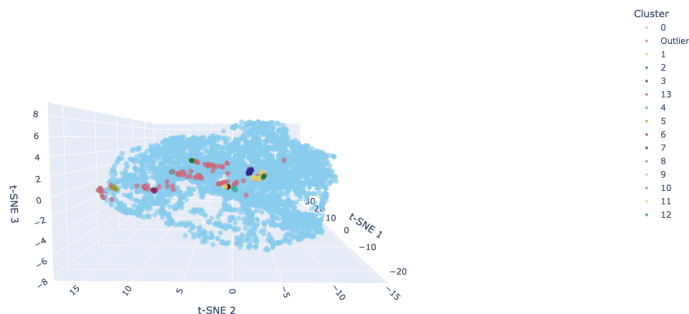
Visualizing with t-SNE

The Challenge

High-Dimensional Complexity

How do we visualize 6-dimensional audio data in a 2D/3D space without losing the underlying structure of the clusters?

DBSCAN Clustering – Interactive 3D t-SNE



The Solution

t-SNE Algorithm

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a non-linear dimensionality reduction technique that excels at preserving local structures in high-dimensional data.

The Outcome

Visual Validation

The resulting 2D map provides a clear visual validation of the clustering results, showing distinct groupings where similar songs are mapped to nearby points in the plane.

Conclusion & Future Work

The Findings

Data-Driven Segmentation

We successfully applied multiple clustering techniques to segment music data. Each algorithm KMeans, DBSCAN, and Agglomerative offered a unique and complementary perspective on the underlying structure of the audio features.

Next Steps

API Integration

Integrate the clustering model with a music player API (e.g., Spotify) to enable real-time playlist generation based on user preferences.

User Validation

Conduct A/B testing with real users to evaluate the perceived accuracy and relevance of the generated music recommendations.

Model Refinement

Explore deep learning embeddings (e.g., VAEs) to capture more nuanced musical relationships beyond standard audio features.

THANK YOU!

Questions & Discussion

Project: Moosic Case Study

